Mayug Maniparambil
email: mayugmaniparambil@gmail.com, ee13b088@ee.iitm.ac.in
website: mayug.github.io
google scholar: https://scholar.google.com/citations?user=tCHi_7IAAAAJ&hl=en

RESEARCH INTERESTS

I am interested in the broad areas of computer vision and machine learning. My specific interests are in making computer vision models learn with limited annotations- semi-supervised learning, few shot learning and weakly supervised learning. I am fascinated by how prior knowledge can be incorporated into vision models to improve performance when annotations are limited.

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institution	%/CGPA	Year of completion
Dual Degree (BTech and MTech) in Electrical Engineering	Indian Institute of Technology Madras, Chennai	8.4	2018
XII	Maharishi International Residential School	94.8	2013
X	Hari Sri Vidya Nidhi School	94.5	2011

- Awarded the **Mitacs Globalink** Research Internship 2016; 1 of 7 students from IIT Madras
- Placed in top 20 percentage of the **Deep Learning** Summer School 2017 IIIT Hyderabad.

COURSE WORK

- Machine Learning
- Deep Learning for Imaging
- Linear Algebra
- Probability and Statistics
- Probabilistic Graphical Models
- Speech Signal Processing
- Image Signal Processing
- Biomedical Signal Processing
- Biomedical Instrumentation
- Analog and Digital Filters
- Digital Communications
- Wireless Communications
- Multi Carrier Communications
- Communication Networks
- Waveguides, MW Circuits & Antenna Design

SKILLS

- Programming Languages- C, C++, Python, Matlab
- Fluent in implementing various neural network architectures using pytorch, tensorflow, luatorch

PUBLICATIONS AND POSTERS

- Paper 'Phase retrieval for Fourier Ptychography under varying amount of measurements' accepted for spotlight presentations at BMVC 2018.
- Poster on Phase retrieval for Fourier Ptychography under varying amount of measurements presented at (Computational Cameras and Displays) CCD workshop, CVPR 2018
- Presented a poster 'Denoising High Density Mouse Brain Images using Deep Generative Models' at Society for Neuroscience Conference 2017 , Washington D.C.
- "Large Scale Topological Analysis of Cranial Bleeds" paper and open source dataset to be submitted to the Lancet (as of Oct 2020).
- 2 Posters presented at machine learning track of Radiological Society of North America (RSNA) Conference 2019 as part of qure.ai Head-CT research team.
    a. AUC and Enriched Datasets are Not Good Enough Anymore: Presenting an Alternative Metric to Evaluate Radiology AI Models
    b. Clinical Context Improves the Performance of AI Models for Cranial Fracture Detection
- 1 Poster presented at machine learning track of European Congress of Radiology (ECR) 2020 on building bleed classification algorithms robust to CT acquisition artifacts.
- IITMSAT Communications Systems-' A Lean Sat Design Approach ' presented at the 3rd IAA Conference on University Satellite Missions and CubeSat Workshop at Rome, Italy.

PROFESSIONAL EXPERIENCE AND PROJECTS

- Data Scientist at Qure.ai — Jun 2018-Oct 2020

  - Responsible for developing cranial bleed subtype detection algorithms-
    - Trained bleed subtype (subdural, epidural, intra-ventricular) detection models weakly supervised by scan-level labels automatically extracted from associated reports using a rule-based NLP algorithm.
    - Used an efficient-net based architecture with shared weights for each slice, followed by a simple differentiable fusion technique to get scan level predictions for each subtype branch. The model at test time gives slice-level predictions for each bleed subtype.
    - This enabled the model to be trained on ~ 1 million Head CT scans each consisting of at least 32 slices without any annotation effort.
    - Measures like composite scaling of depth, width and resolution of networks, auto augmentation search, implementing squeeze-excitation blocks etc. lead to significant improvements in tracked metrics. (AUC, AP)
    - This work is part of FDA approved product qER which is deployed in over 10 countries

  - Responsible for developing cranial bleed segmentation, quantification algorithms-
    - Gated Attention UNet model trained with focal loss to predict bleed masks and regress the volume of bleed
    - Implemented a novel multi-stage training process making use of both coarse and fine bleed location segmentation maps.
    - This work led to the product qQuant which has been deployed in over 7 countries.

  - Active learning based 3d annotation portal-
    - Designed and developed an active learning-based framework for cranial bleed segmentation annotations and lateral ventricle segmentation annotations.
    - The framework automatically identified low performing strata to pick new scans for the annotators. This was important to quickly improve performance in low performing strata.
    - Initially trained segmentation models were used to prepopulate the scans with abnormality masks to speed up the annotation process. This reduced the annotation time from ~40 mins per scans to less than 10 mins.
    - Designed and developed a web annotation portal with a 3d-slicer backend that follows the above-mentioned active-learning framework.

  - Large Scale Analysis of Cranial Bleed Topologies
    - Designed a framework and dataset for analyzing bleed topologies in a set of 1 million patients. For instance, using this dataset, one can visualize the bleed topologies of male stroke patients aged 30-40 years as well as test hypothesis of the form "Is bleed prevalence more in the left temporal region for this subset? "
    - This was achieved by running a bleed segmentation algorithm on all of 1 million scans, followed by CT registration, and parcellation algorithms to get a table with information about abnormalities, bleed coverages in each parcel, demographic and clinical characteristics for each patient.
    - This work is the first large scale analysis, and dataset for cranial bleeds and their topologies and is set to be published in a high impact radiology journal (Lancet).

  - Cranial fracture segmentation algorithm and fracture location reporting-
    - Used a UNet based model to segment out bones in cranial scans and used the segmentation maps in conjunction with predictions from a fracture segmentation model to report the fracture location.

- Phase Retrieval for Fourier Ptychography (FP) Microscopy (Dual Degree Project under Prof. Mitra) — Feb-May 2018

  - Developed an auto-encoder based architecture that can be adapted for performing FP phase retrieval under varying levels of constraints by using same generator network, but different optimization frameworks depending on the amount of constraints.
  - For low constraints case, we proposed supervised learning-based technique that learns a conditional prior to map the low-resolution measurements to its high-resolution phase and amplitude.
  - For higher constraints, we proposed a novel unsupervised framework, that optimize over the generator parameters by minimizing the forward measurement error of FP. The proposed method exploits the low-level image statistics captured by generator network's inherent structure, thereby making it more robust to phase-amplitude leakage.
  - Using simulations for uncorrelated phase and amplitude in both low and high constraints, the proposed algorithm outperformed previously proposed FP phase retrieval

- Medical Image Denoising (Dual Degree Project under Prof. Kaushik Mitra) Jun-Dec 2017

  - Developed a GANs based Generative model to inpaint serial two photon tomography images of mouse brain in collaboration with Dr. Pavel Osten's Lab, Cold Spring Harbor Laboratory. Proposed an encoder-decoder network with adversarial loss to remove high expression regions in mice brain by inpainting that matches the context.

  - Gained a solid understanding of various generative models (GANs, variational autoencoders, auto regressive models)

- Wireless RF tag for Volatile Gas Sensing under Prof. Gregory Bridges at University of Manitoba, CN Summer 2016
  - Designed and simulated an active feedback Microwave resonator to improve the quality factor of microwave sensors
  - Implemented and tested the circuit on FR4 board.

- Source Separation Final Project of Speech Signal Processing Course Winter 2016
  - Used Robust PCA based method to separate vocals from instrumental music in music tracks with remarkable perceptible quality.

- Affordable Gait Visualization System Final Project of Biomechanics Course Winter 2014
  - Used IMU data from old smartphones, and Arduino to create a Gait Visualization system using Python and Blender

- IIT Madras Satellite Project- Onboard Communications 2015-2016
  - Interface circuit of ext. power amp for ADF7023 transceiver board.
  - RF Routing within the satellite structure

TEACHING

- **Teaching Assistant** for Advanced Microelectronics Lab for dual degree Students Aug-Nov 2017
  - Conducted Lab Sessions twice a week, which involved explaining the experiments and evaluating lab manuals.
  - Helped in setting and evaluating the end semester paper.

- **Teaching Assistant** for Image Processing EE 5175 for dual degree Students Jan-May 2018
  - Set tutorial questions and conducted tutorial sessions every week
  - Helped in evaluation of quiz papers and programming assignments

REFERENCES

- Prof. Kaushik Mitra
  Dual degree thesis supervisor,
  Assistant Professor, Electrical Engineering Dept., IIT Madras.
  http://www.ee.iitm.ac.in/kmitra/
  kmitra@ee.iitm.ac.in

- Dr. Prashant Warier
  Co-founder and CEO,
  Qure.ai
  https://www.linkedin.com/in/pwarier/
  prashant.warier@qure.ai

- Dr. Pooja Rao
  Head R&D
  Qure.ai
  https://nl.linkedin.com/in/raopooja
  pooja.rao@qure.ai

- Prof. Gregory Bridges
  Mitacs Globalink internship supervisor
  Professor, Electrical Engineering Dept., University of Manitoba
  Gregory.Bridges@umanitoba.ca